## SECTION B: CHECKSUMS, ECHO CHECKS & CHECK DIGITS (10 MARKS)

**Q3. Checksum Verification Process** *[3 Marks]*

Describe how a checksum algorithm ensures that a transmitted data file has arrived without corruption.

**Q4. Echo Check Critique** *[3 Marks]*

A student proposes using an Echo Check for online multiplayer gaming. Give TWO distinct reasons why an Echo Check is unsuitable for high-speed network applications.

Reason 1:

Reason 2:

**Q5. Check Digits vs Transmission Checks** *[4 Marks]*

(a) State TWO human data entry errors that check digits detect but parity checks do not. *[2]*

(b) The following 12 digits of an ISBN-13 barcode are scanned: `9 7 8 1 5 1 0 4 8 4 4 2`.
Calculate the final 13th Check Digit using the weighting `(1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3)`. Show your calculation: *[2]*